

Term 7- Sept 2025

Nanoelectronics and Technology
(01.119/99.503)-Week 5 Class 1

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14-Oct- 2025

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Outline

- IC Fabrication
 - Ion Implantation
 - Diffusion
 - Thin film deposition techniques
 - Photo lithography

Mask and its Parameters

- Quartz material of choice for steppers.
 - very low coefficient of thermal expansion
 - High light transmission in UV and DUV wavelengths.

The term “**steppers**” refers to **photolithography steppers** — highly precise optical projection systems used in **semiconductor manufacturing**.

A **stepper** (short for *step-and-repeat photolithography system*) is a **machine used to project circuit patterns** from a photomask onto a silicon wafer during IC fabrication.

It's called a *stepper* because it:

- 1.**Projects** the image of a small area (die) of the mask onto the wafer.
- 2.Then **steps** the wafer to the next position.
- 3.**Repeats** the exposure process across the entire wafer — hence “step-and-repeat.”

This process is a core part of **photolithography**, which defines the fine patterns of transistors and interconnects on chips.

CMOS Fabrication Steps

- Trench isolation to define active regions
- N well definition for PMOS transistors
- P well definition for NMOS transistors
- Gate oxide formation and gate poly silicon patterning
- NMOS halo and shallow extension implant
- PMOS halo and shallow extension implant
- Spacer formation
- N+ select for NMOS source/drain
- P+ select for PMOS source/drain
- S/D activation and Silicidation
- Contact definition
- Metal 1 layer patterning

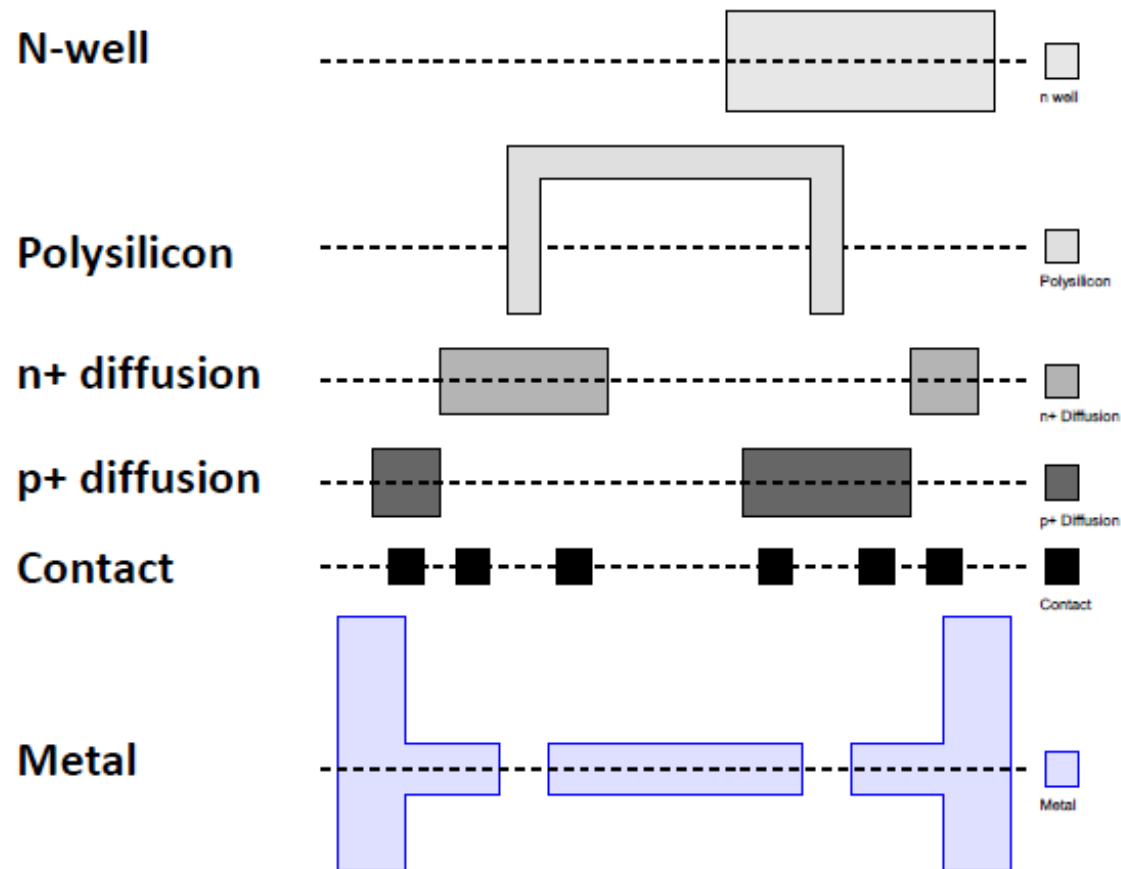
Ref: N. Bhat, CeNSE, IISc Bangalore- Nanoelectronic Device Technology

CMOS Fabrication Steps

- CMOS transistors are fabricated on silicon wafers
- Lithography process has been the mainstream chip manufacturing process
- On each step, different materials are deposited or etched
- Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process

CMOS Fabrication Steps

Six masks to build simple inverter



IC manufacturing- FEOL and BEOL

• FEOL Processes

- Si epitaxy
- Ion implant / diffusion
- Oxidation
- Photolithography /Etch
- RTO

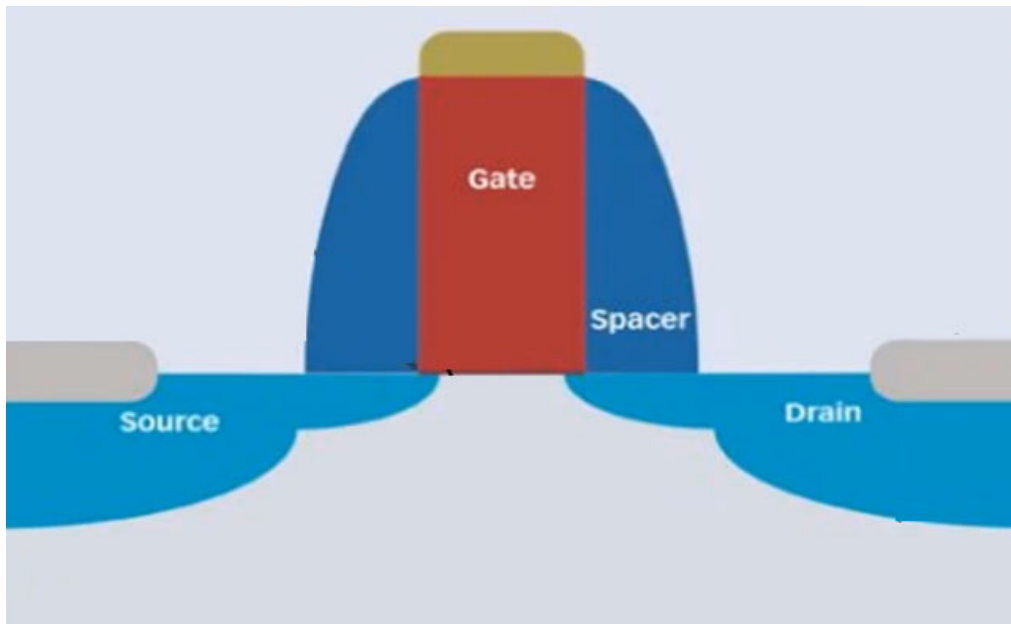
Create the transistors

• BEOL Processes

- PVD/sputtering
- CVD
- PECVD
- Plasma etching
- Photolithography
- CMP
- Electroplating

Do the wiring

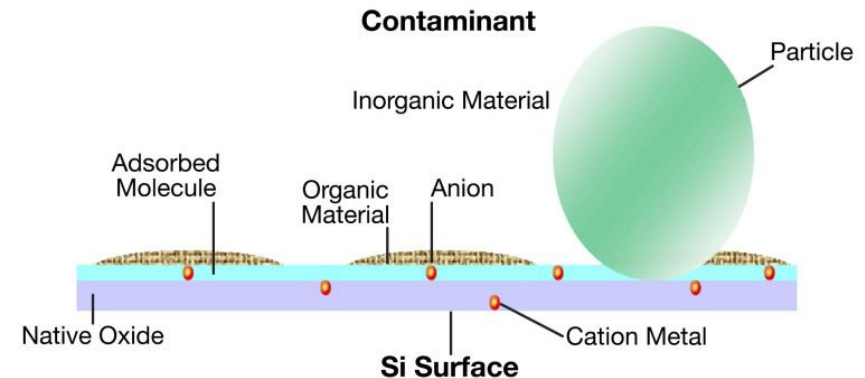
MOSFET Fabrication



- Silicon substrate preparation-
RCA cleaning
- Doping –Ion Implantation
- Photolithography-
Mask+Photoresist+Etching
- Deposition
- Contacts
- Cleaning

Silicon substrate preparation

- RCA cleaning refers to a two-step cleaning process for silicon wafers used in semiconductor manufacturing to remove organic and inorganic contaminants.
- It involves using chemical solutions (SC-1 and SC-2) at high temperatures and was developed by the Radio Corporation of America (RCA) in 1965. The process is critical for ensuring the purity and integrity of wafers before high-temperature steps like oxidation and diffusion.



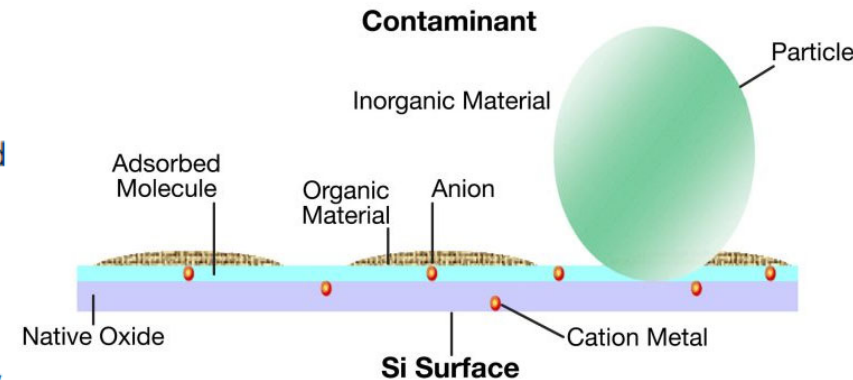
<https://www.mksinst.com/n/wafer-surface-cleaning>

https://en.wikipedia.org/wiki/RCA_clean

Silicon substrate preparation

The RCA cleaning process

- **Step 1 (SC-1):** Removes organic contaminants. The wafer is submerged in a heated solution of ammonium hydroxide (NH_4OH), hydrogen peroxide (H_2O_2), and deionized water.
- **Step 2 (SC-2):** Removes inorganic contaminants and metallic ions. The wafer is placed in a heated mixture of hydrochloric acid (HCl), hydrogen peroxide (H_2O_2), and deionized water.
- **Final rinse and dry:** After the chemical baths, the wafer is rinsed with deionized water and dried. Some variations of the process may also include a final dip in hydrofluoric acid (HF) to remove the native oxide layer.



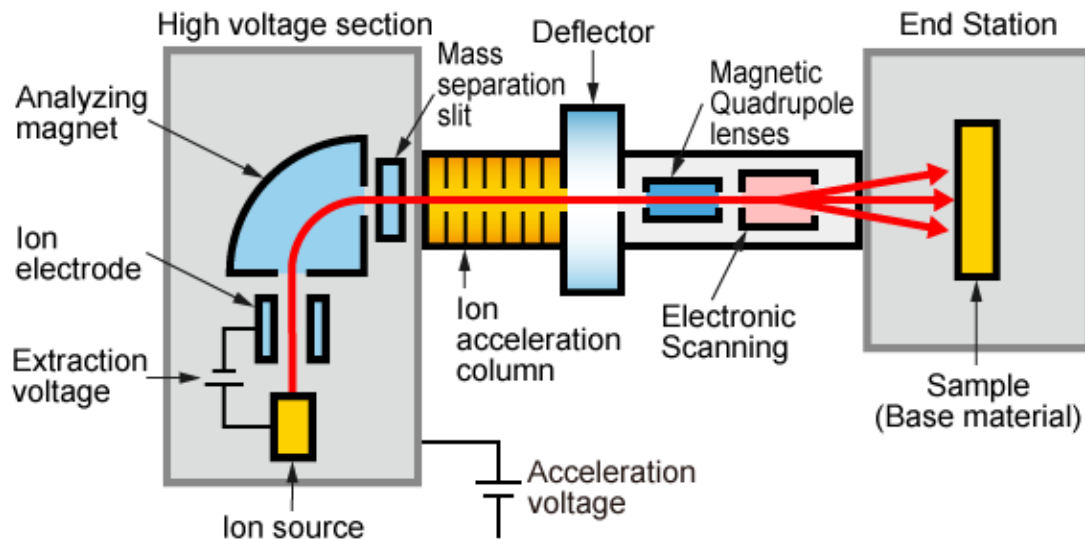
<https://www.mksinst.com/n/wafer-surface-cleaning>

1. Removal of the organic contaminants (organic clean + particle clean)
2. Removal of thin oxide layer (oxide strip, optional)
3. Removal of ionic contamination (ionic clean)
4. Helps maintain the electrical characteristics and reliability of the final product.

https://en.wikipedia.org/wiki/RCA_clean

Ion Implantation

- **Ion implantation** is a process used to **introduce controlled amounts of dopant atoms** (e.g., B, P, As) into a semiconductor wafer.
- A **beam of high-energy ions** is directed at the wafer → ions penetrate the surface and embed at controlled depths.
- It is one of the most **precise methods of doping**, replacing older diffusion methods in advanced IC fabrication



Main Application:

Doping (S/D or n/p well) in CMOS

Ref: <http://www.cityu.edu.hk/phy/appkchu/AP6120/9.PDF>

Ion Implantation -Doping

- Ion implantation is a **non-thermal doping process** where ions of dopant atoms are **accelerated by an electric field** and **bombarded into the silicon wafer**.
- In ion implantation, dopant atoms are volatilized, ionized, accelerated, separated and directed at a target that is typically a silicon substrate. The atoms enter the crystal lattice, collide with the host atoms, lose energy, and finally come to rest at some depth within the solid.
- The average penetration depth is determined by the dopant, substrate materials, and acceleration energy.
- Ion implantation energies range from several hundred to several million electron volts, resulting in ion distributions with average depths from < 10 nm to $10\text{ }\mu\text{m}$. Doses range from 10^{11} atoms/cm² for threshold adjustment to 10^{18} atoms/cm² for buried dielectric formation.

Ion Implantation -Doping

Process Steps:

- 1. Ion Generation:** Dopant gas (e.g., B_2H_6 , PH_3 , AsH_3) is ionized in a plasma chamber.
- 2. Acceleration:** The ions are accelerated to high energy (10 – 500 keV).
- 3. Beam Focusing and Scanning:** A magnetic/electrostatic field directs the ion beam onto the wafer surface.
- 4. Implantation:** Ions penetrate the crystal lattice and come to rest at specific depths depending on their energy.
- 5. Annealing:** A rapid thermal anneal (RTA) at 800 – 1050 °C repairs lattice damage and activates the dopants (puts them on substitutional sites).

Ion Implantation -Doping

Ion Beam Current (I)

The **beam current** represents the **number of ions per second** striking the wafer and controls the **dose rate** (how fast dopants are implanted).

Application	Beam Current (I)	Remarks
Low-current beamline implanter	1 μ A – 1 mA	High precision, low contamination
High-current implanter	1 – 10 mA	For high-dose implants (e.g., source/drain)
High-energy implanter	10 μ A – 1 mA	For deep wells or buried layers

Ion Implantation -Doping

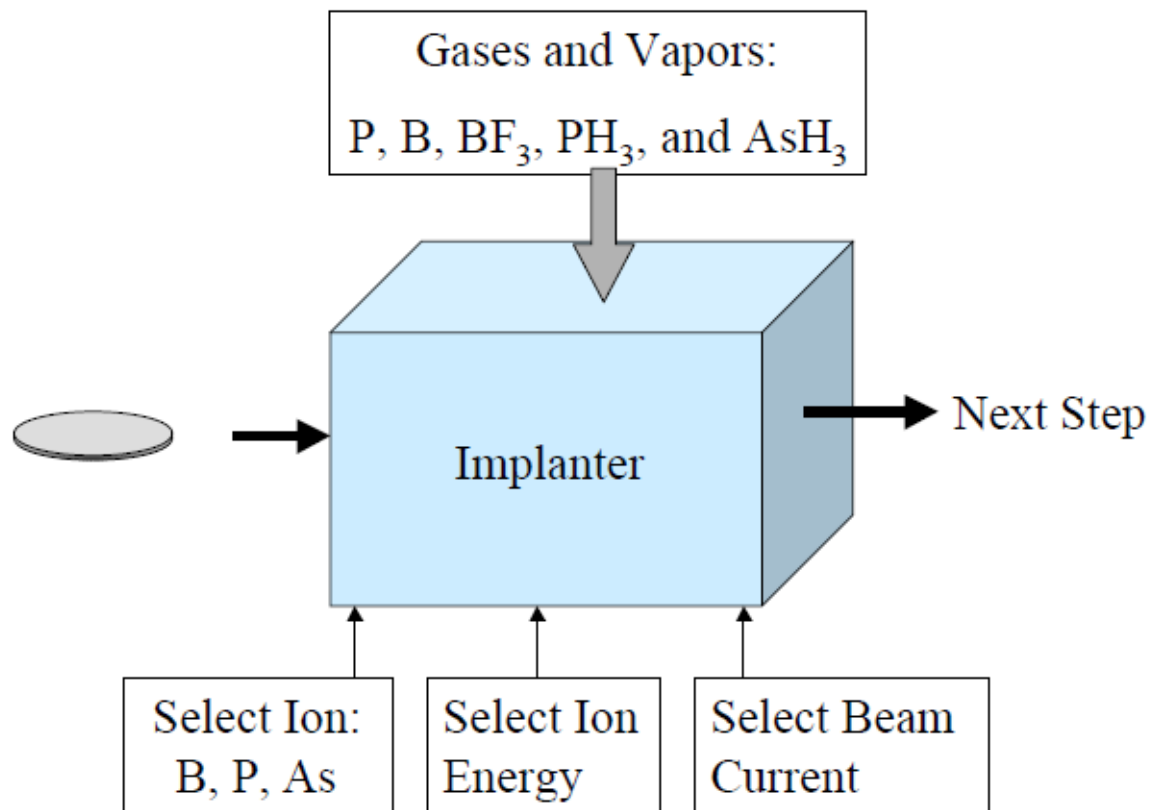
Ion Beam Energy (E)

The **ion energy** determines **how deep** the dopant ions penetrate into the semiconductor.

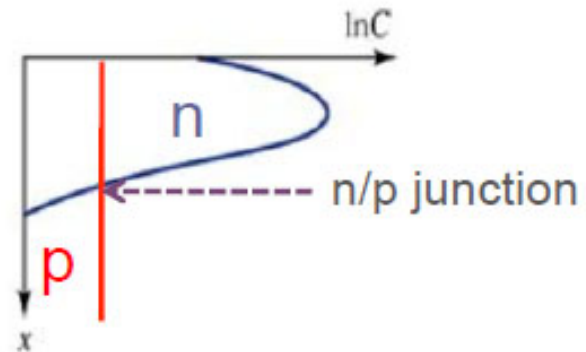
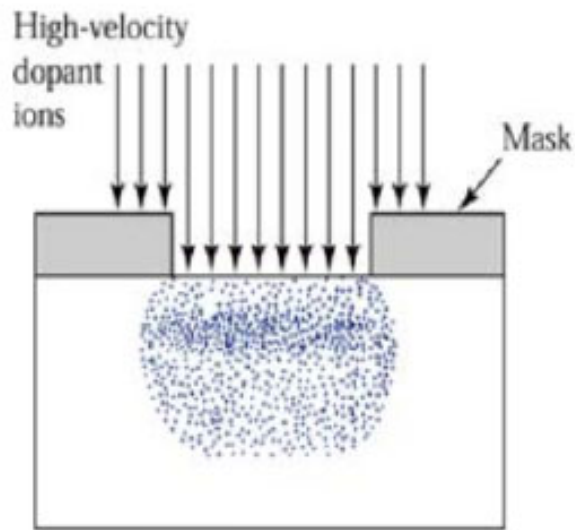
Dopant	Typical Ion Energy	Typical Depth (Range)
Boron (B^+)	10 – 50 keV	0.05 – 0.3 μm
Phosphorus (P^+)	50 – 200 keV	0.1 – 0.5 μm
Arsenic (As^+)	100 – 300 keV	0.05 – 0.5 μm

Ion Implantation -Doping

Implantation Process



Ion Implantation -Doping



ion implantation

G. Lulli – CNR – Institute for Microelectronics and Microsystems (IMM) - Bologna

Ion Implantation -Doping

Key Parameters:

- **Dose (ions/cm²):** Controls dopant concentration.
- **Energy (keV):** Controls penetration depth.
- **Angle:** Affects channeling (ions traveling along crystal directions).
- **Temperature:** Implantation often done at room temperature; annealing follows later.

Advantages:

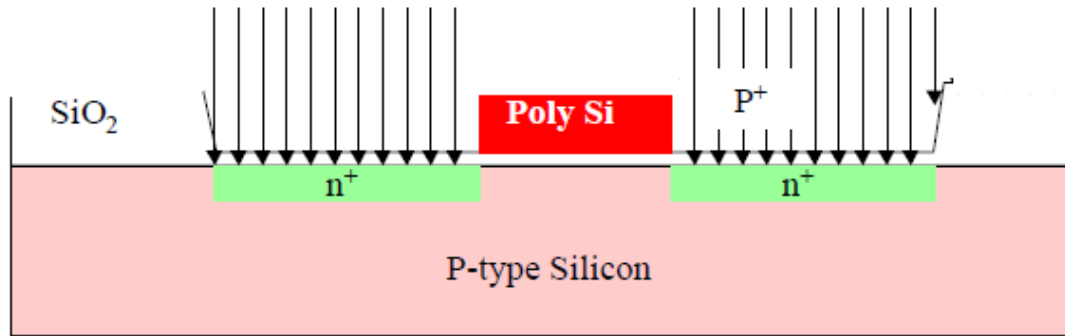
- Precise control of dose and depth.
- Low temperature process.
- Can create shallow junctions for modern CMOS (nm-scale).

Ion Implantation -Doping

- Dope semiconductor
- Better doping method than diffusion
- Easy to control junction depth (by ion energy) and dopant concentration (by ion current and implantation time).
- Anisotropic dopant profile.

www2.austin.cc.tx.us/HongXiao/Book.htm

Ion Implantation -Doping

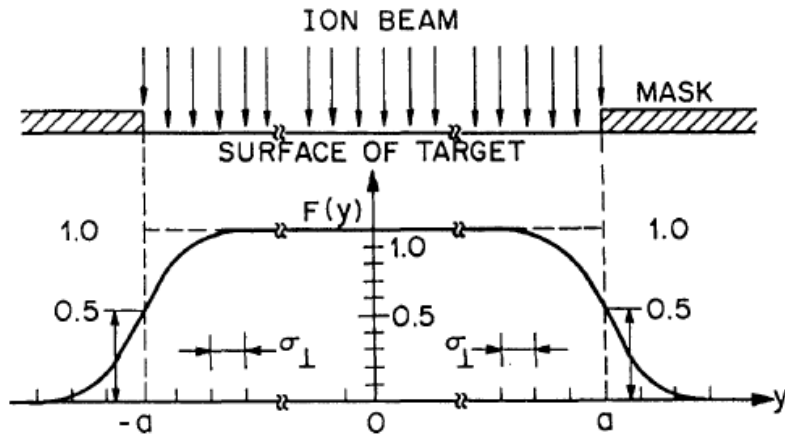


Ion Implantation -Doping



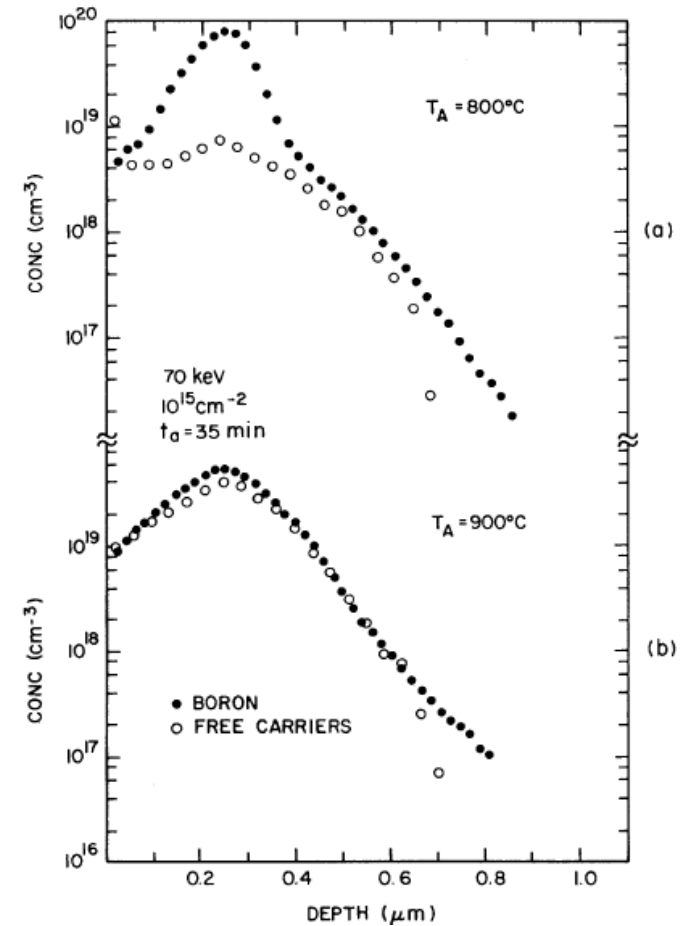
<https://www.youtube.com/watch?v=8FM7JSap9WA>

Ion Implantation -Doping



Two-dimensional implantation profile showing fraction of total dose as a function of lateral position for an opaque mask.

Ref: <http://www.cityu.edu.hk/phy/appkchu/AP6120/9.PDF>



Concentration profiles of boron atoms and corresponding free carrier concentrations

Ion Implantation –Doping

- P-Type: Boron- Light element=> Diffuses faster
- Indium- Heavy element=> Diffuses slower
- N-Type: Phosphorus- Light element=> Diffuses faster
- Antimony Heavy element=> Diffuses slower
- Shallow Source and Drain: Heavy element preferred.

Doping-Diffusion

- **Definition:**

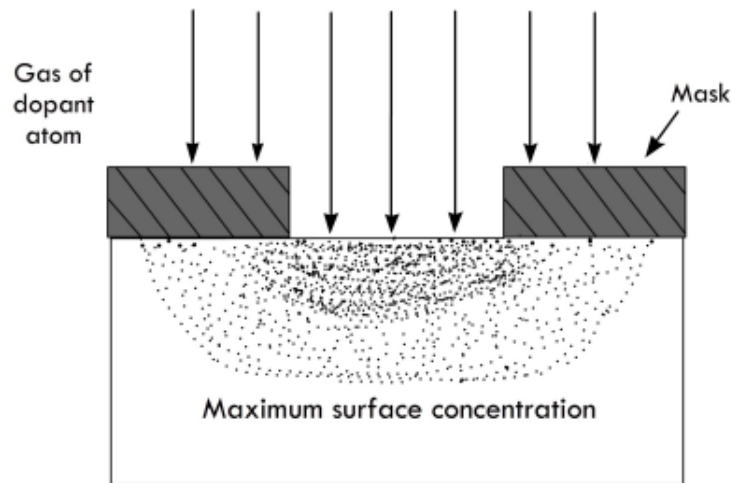
Diffusion is a **thermal process** in which dopant atoms move into silicon at high temperature due to **concentration gradients**.

- Diffusion doping is a semiconductor manufacturing process where dopant atoms move from an area of high concentration to one of lower concentration into a semiconductor wafer.
- This is achieved by heating the wafer in a furnace with dopant atoms, typically from a gas or liquid source, which then move through the crystal lattice and change the semiconductor's electrical conductivity.

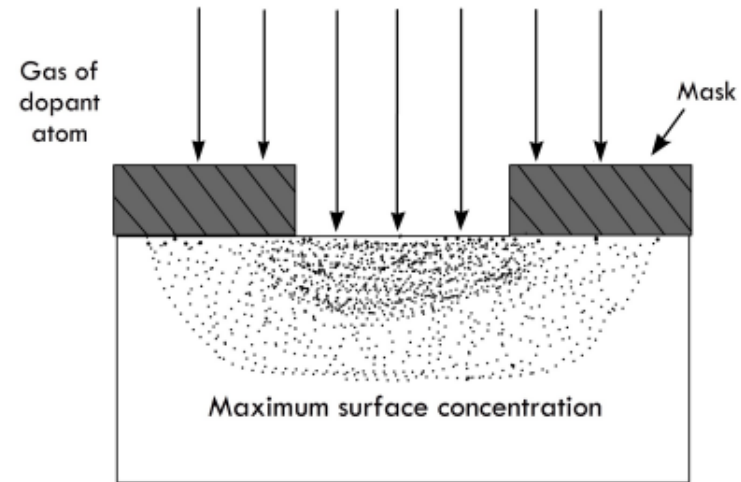
Doping-Diffusion

Thermal diffusion is a two step process, similar to the steps in oxidation process by consuming the underlying Si.

1. Deposition - dopant atoms are introduced at the wafer surface.
2. Drive-in - the dopant atoms then diffuse into the wafer to create the required concentration gradient.



Doping-Diffusion



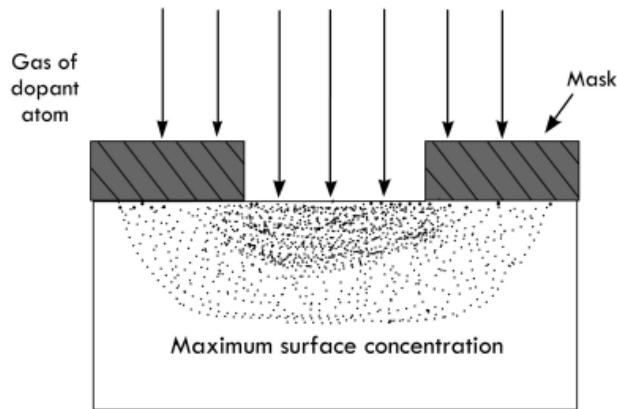
Fick's First Law (Steady State)

$$J = -D \frac{dC}{dx}$$

Where:

- J = diffusion flux (atoms/cm²·s)
- D = diffusion coefficient (cm²/s)
- C = dopant concentration (atoms/cm³)
- x = distance into the wafer

Doping-Diffusion

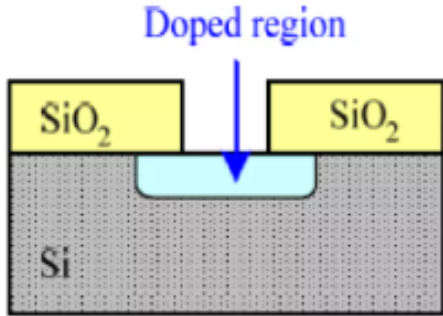
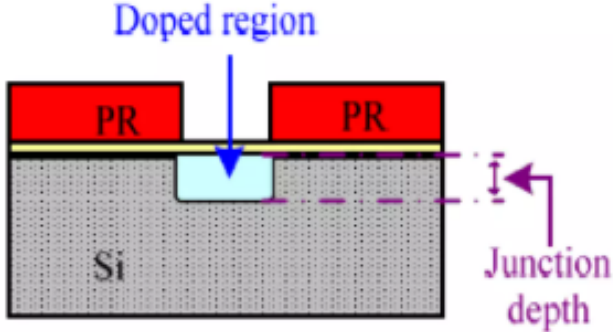


Factors Affecting Diffusion

Factor	Effect
Temperature	$\uparrow T \rightarrow \uparrow D$ exponentially (Arrhenius law)
Time	Longer time $\rightarrow$ deeper diffusion
Dopant type	Lighter atoms (B) diffuse faster than heavier ones (As)
Crystal defects	Vacancies enhance diffusion rate
Ambient gas	O_2 , N_2 , or inert gases influence surface reactions

Doping-Diffusion

Comparison

Diffusion	Ion Implantation
♦High temperature, hard mask (SiO_2) ♦Isotropic dopant profile ♦Cannot independently control the dopant concentration, junction depth	♦Low temperature, PR mask ♦Anisotropic dopant profile ♦Can independently control the dopant concentration, junction depth
 The diagram shows a cross-section of a silicon (Si) substrate. A yellow layer of silicon dioxide (SiO_2) covers the top surface. A blue arrow labeled 'Doped region' points down into a rectangular opening in the SiO_2 layer, indicating the diffusion of dopant into the Si substrate. The dopant profile is shown as a broad, isotropic region.	 The diagram shows a cross-section of a silicon (Si) substrate. A red layer of photoresist (PR) covers the top surface. A blue arrow labeled 'Doped region' points down into a rectangular opening in the PR layer, indicating the implantation of dopant into the Si substrate. The dopant profile is shown as a narrow, anisotropic region. A purple dashed line with a double-headed arrow indicates the 'Junction depth'.

Doping-Diffusion

Comparison of Implantation and Diffusion

Diffusion	Ion Implantation
High temperature, hard mask	Low temperature, photoresist mask
Isotropic dopant profile	Anisotropic dopant profile
Cannot independently control of the dopant concentration and junction depth	Can independently control of the dopant concentration and junction depth
Batch process	Both Batch and single wafer process

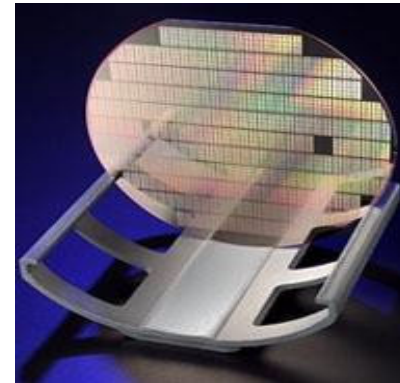
Factors Effecting Ion Implantation

- Beam current (dosage)
- Vacuum purity (contamination)
- Beam energy (depth of implant)



Factors Effecting Diffusion

- Time
- Type of Dopant
- Temperature

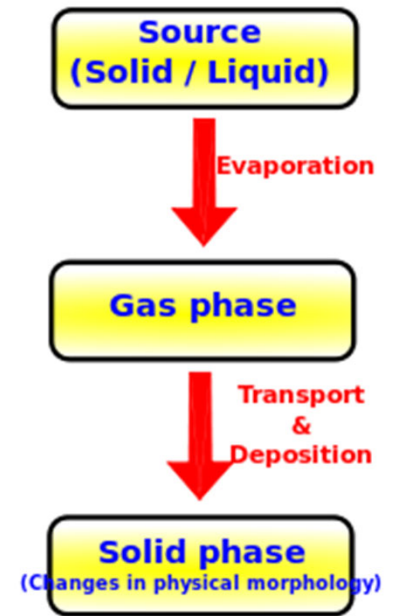


Deposition Techniques

- **Physical Vapour Deposition (PVD)**
- **Chemical Vapor Deposition (CVD)**
- **Atomic Layer Deposition (ALD)**

Physical Vapor Deposition

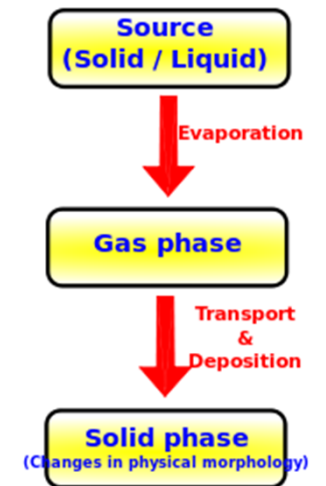
- Physical vapor deposition (PVD) is a **process used to produce a metal vapor that can be deposited on electrically conductive materials** as a thin, highly adhered pure metal or alloy coating. The process is carried out in a vacuum chamber at high vacuum
- PVD is characterized by a process in which the material goes from a condensed phase to a vapor phase and then back to a thin film condensed phase.
- The most common PVD processes are sputtering and evaporation



https://en.wikipedia.org/wiki/Physical_vapor_deposition

Physical Vapor Deposition

HAUZER

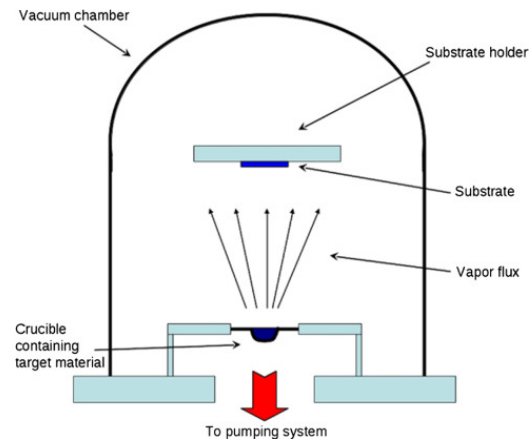


<https://www.youtube.com/watch?v=TFCATQ3VbFY&t=0>

https://en.wikipedia.org/wiki/Physical_vapor_deposition

Thermal evaporation PVD

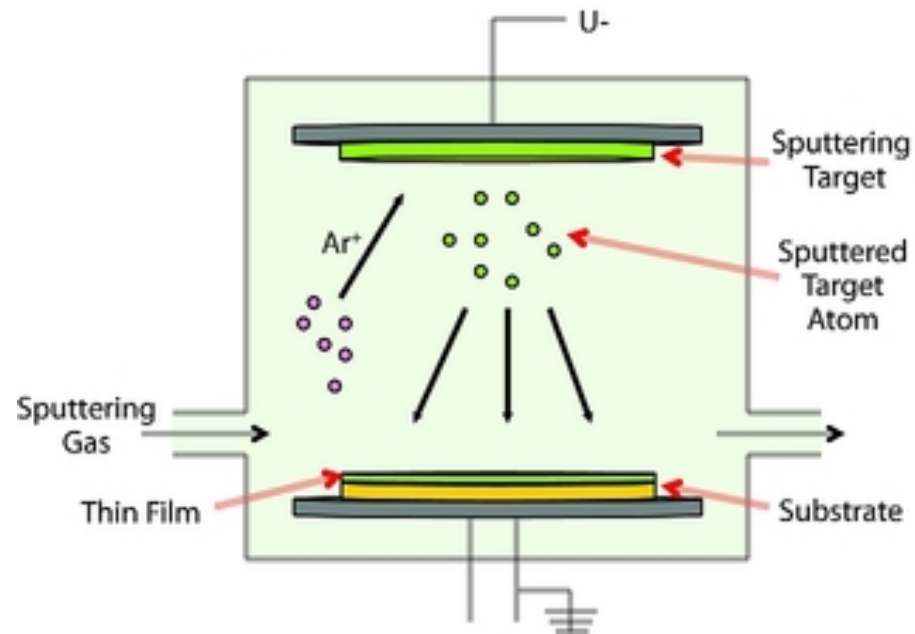
- Thermal evaporation is a common method of PVD.
- It is one of the simplest forms of PVD and typically uses a resistive heat source to evaporate a solid material in a vacuum environment to form a thin film.
- The material is heated in a high vacuum chamber until vapor pressure is produced.



<https://www.sciencedirect.com/topics/chemistry/thermal-evaporation>

Sputtering-PVD

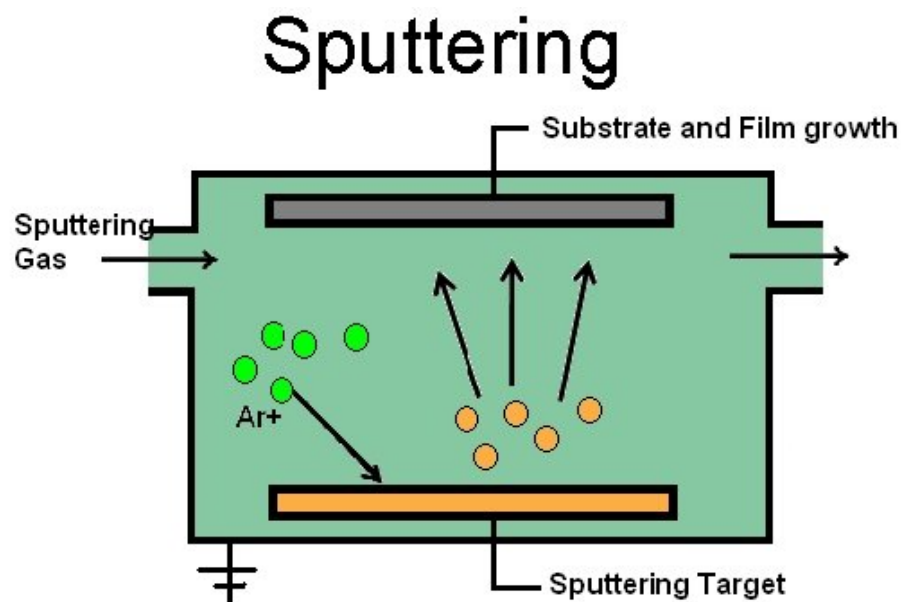
- Sputter deposition is a physical vapor deposition (PVD) method of depositing thin films by sputtering.
- Ejecting material from a target or source, which then deposits onto a substrate – for example, a silicon wafer.



<https://www.sputtertargets.net/blog/advantages-of-physical-vapor-deposition.html>

Sputtering PVD

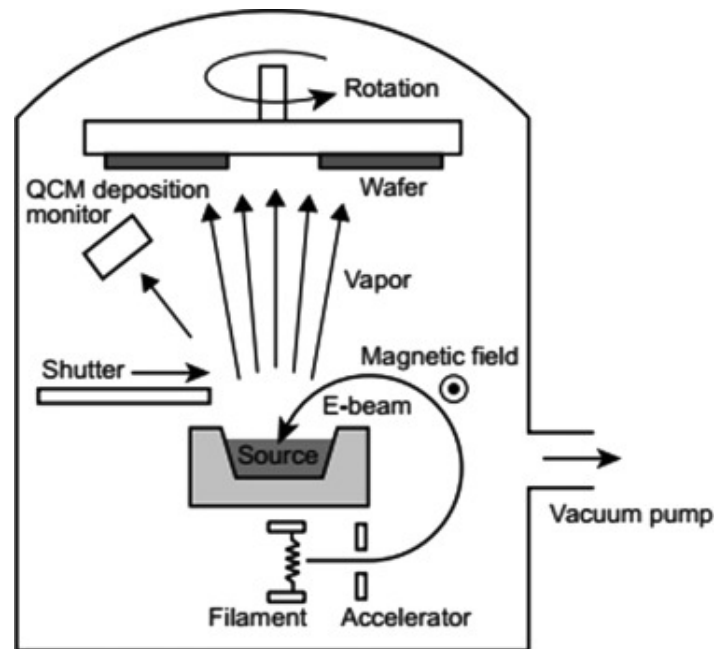
- Highly energized Ar^+ ions directed at a target
- Ar^+ ions sputter target atoms
- Ejected atoms are transported to the substrate
- Atoms condense and form a thin film on the substrate



<https://slidetodoc.com/uses-of-plasmas-in-deposition-sputtering-outline-additive/>

PVD: e-beam evaporation

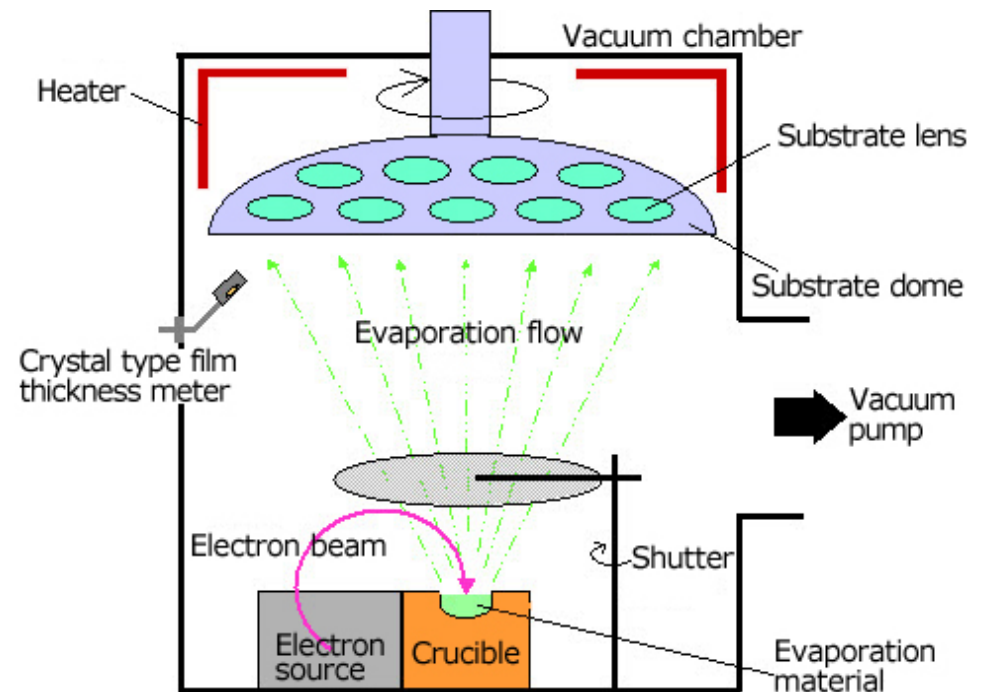
- Electron beam-physical vapor deposition (EB-PVD) utilizes an electron gun to vaporize the coating material.
- To allow the passage of electrons to the evaporation material, the deposition chamber must be under vacuum condition



<https://www.sciencedirect.com/topics/chemistry/electron-beam-evaporation>

PVD: e-beam evaporation

- E-beam is generated by heating a filament above 2500°C using electrical current
- High energy electrons leave the heated filament and focused to crucible using magnets
- Evaporated atoms flow towards the substrate and form a thin film



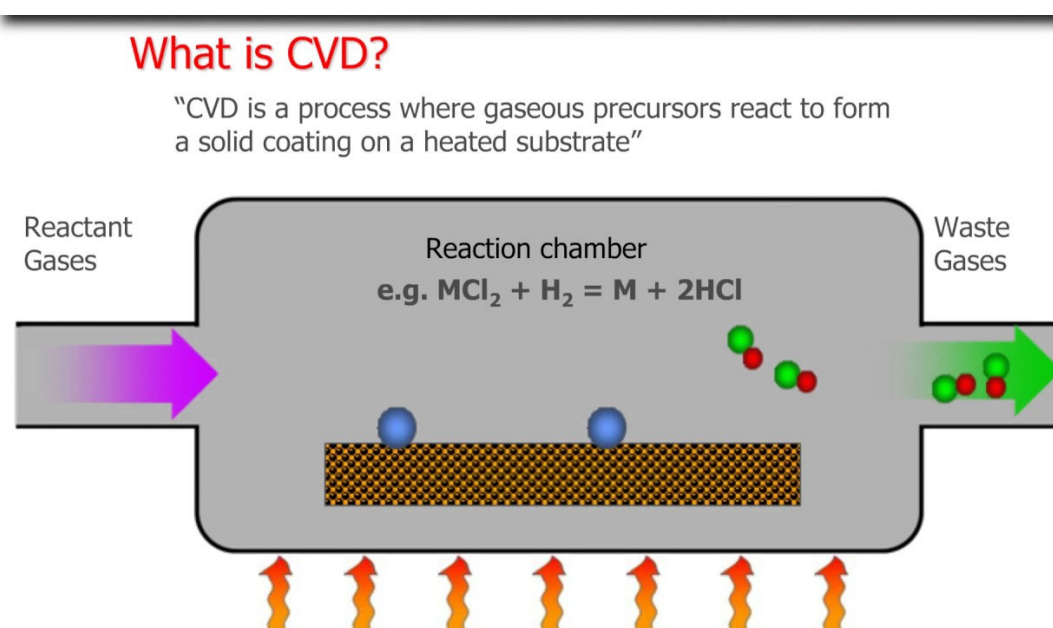
Ref: <https://www.jeol.co.jp/en/science/eb.html>

Physical Vapour Deposition

- Advantages:
 - Low deposition temperature
 - Elemental, alloy, compound (wide variety of materials) coating possible
- Disadvantages:
 - Poor Step Coverage on Complex Geometries
 - Lower coating rate
 - Vacuum condition requirement
 - Sometimes **adhesion layers (e.g., Ti, Cr)** are needed to improve bonding
 - Film thickness can vary across large substrates or wafer batches.

Chemical Vapor Deposition

- Chemical vapor deposition (CVD) is a coating process that uses thermally induced chemical reactions at the surface of a heated substrate, with reagents supplied in gaseous form.



Chemical Vapour Deposition

- CVD is a vacuum deposition method used to produce high quality, and high-performance, solid materials. The process is often used in the semiconductor industry to produce thin films.
- In typical CVD, the wafer (substrate) is exposed to one or more volatile precursors, which react and/or decompose on the substrate surface to produce the desired deposit. Frequently, volatile by-products are also produced, which are removed by gas flow through the reaction chamber.

https://en.wikipedia.org/wiki/Chemical_vapor_deposition

Chemical Vapour Deposition



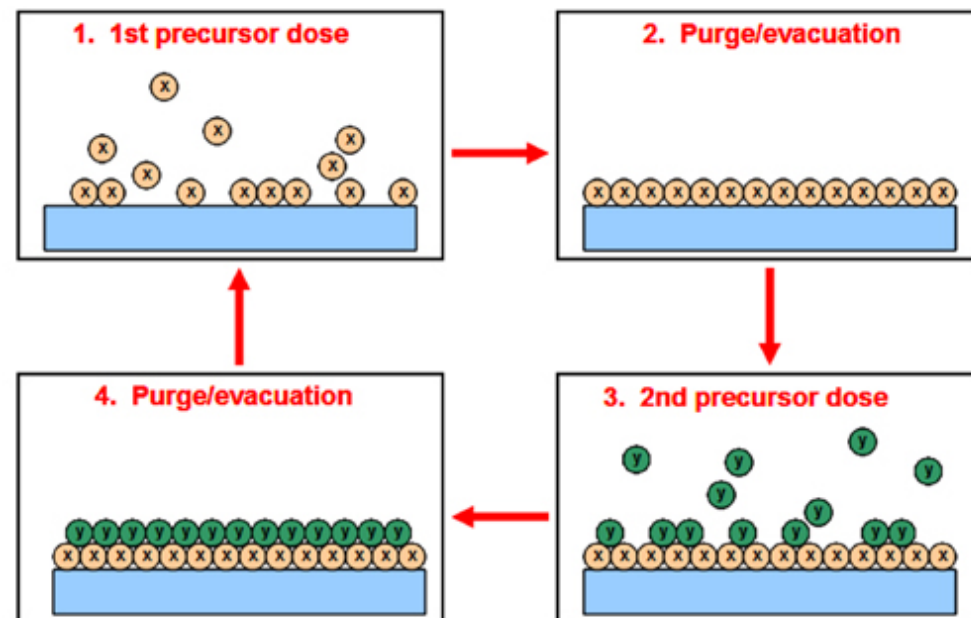
<https://www.youtube.com/watch?v=TFCATQ3VbFY&t=0>

Chemical Vapour Deposition

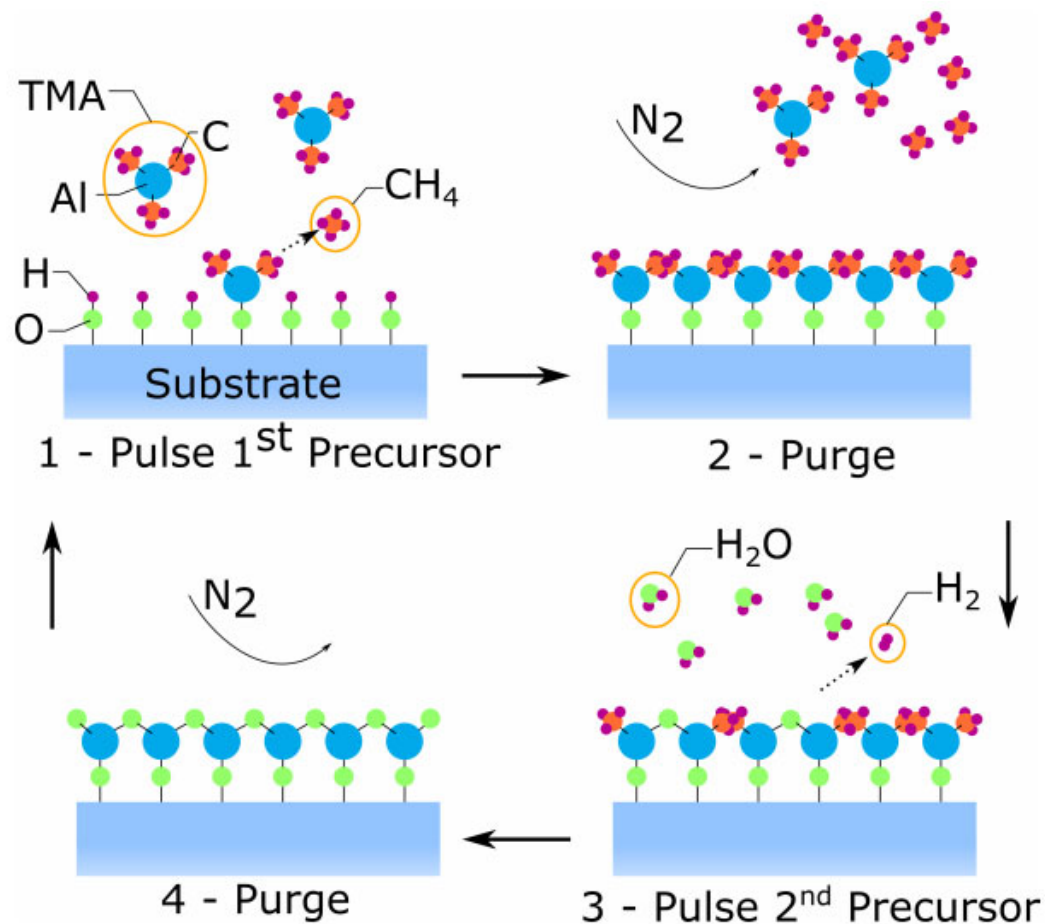
- Advantages:
 - Higher quality deposition and better growth rate
 - Good reproducibility
 - Suitable for thin film
- Disadvantages:
 - Longer reaction time
 - High Processing Temperature
 - Vacuum condition requirement
 - CVD setups demand complex gas-handling systems and strict safety protocols.
 - Particle Formation and Contamination

Atomic Layer Deposition (ALD)

- **ALD** is a thin-film deposition technique based on the sequential use of a gas-phase chemical process. The majority of ALD reactions use two chemicals called precursors (also called "reactants").
- These precursors react with the surface of a material one at a time in a sequential, self-limiting, manner. A thin film is slowly deposited through repeated exposure to separate precursors.



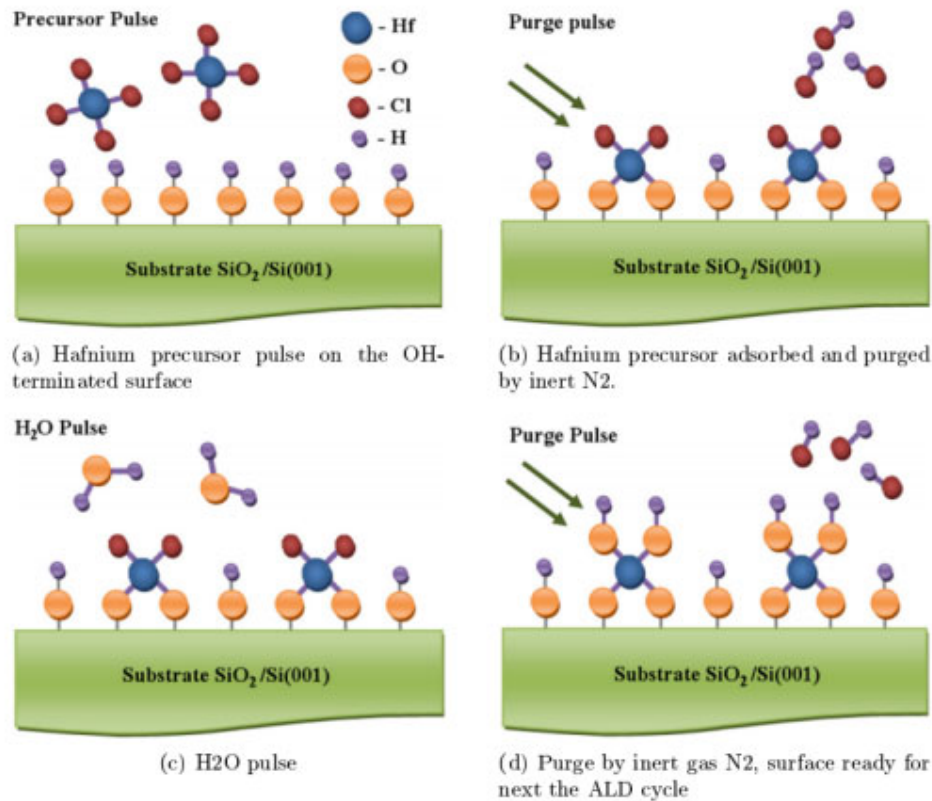
Atomic Layer Deposition: Deposition of Al_2O_3



- The layer thickness can be controlled by number of cycles

https://mms.is.mpg.de/research_projects/atomic-layer-deposition

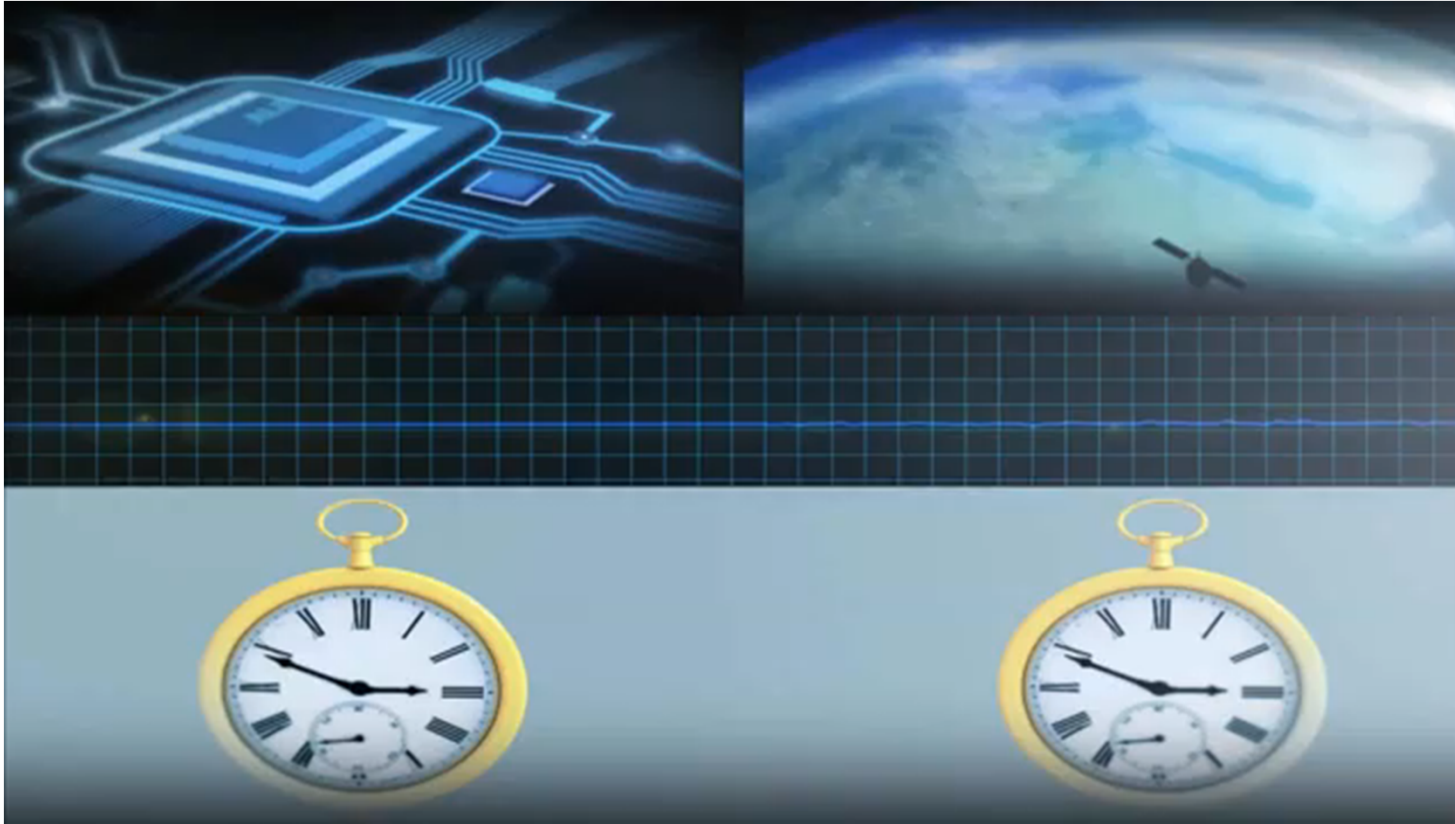
Atomic Layer Deposition: Deposition of HfO₂



Schematic illustration of an ALD cycle of a Hf-oxide process where HfCl₄ precursors and H₂O are alternately pulsed and separated by inert N₂ gas pulsing

- The layer thickness can be controlled by number of cycles

Atomic Layer Deposition



<https://www.youtube.com/watch?v=CvkF-trrWvA>

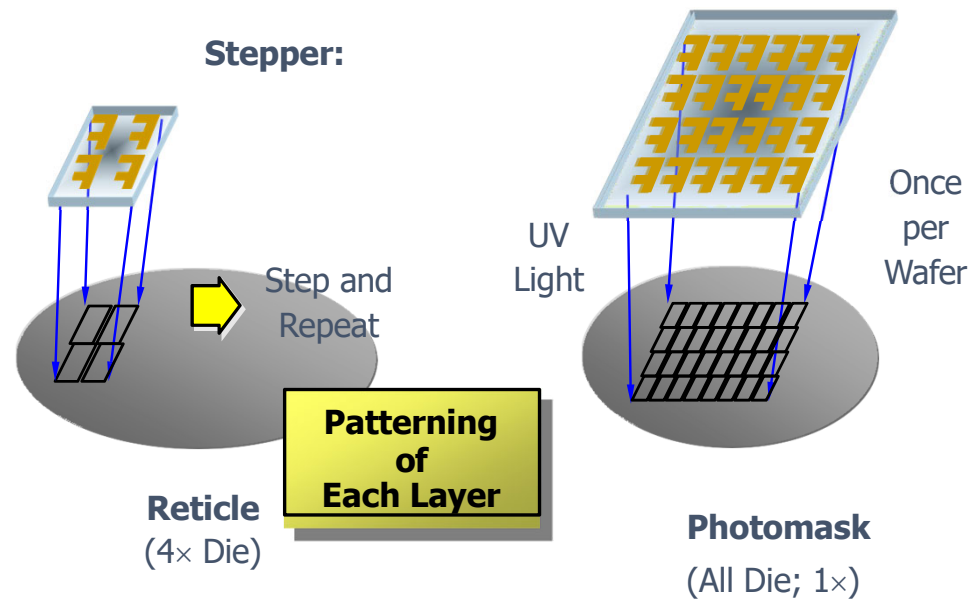
Atomic Layer Deposition



Atomic Layer Deposition

- Advantages:
 - Very high film quality
 - Better film thickness control
 - Good conformability
 - Suitable for dielectric thin film deposition
- Disadvantages:
 - Slow deposition rate-ALD deposits one atomic layer per cycle, typically 0.1–0.2 nm/cycle.
 - More complicated reactor designs and more operating cost
 - Limited Choice of Suitable Precursors

Masking Transfers Patterns



- A **reticle** is more precise but involves step-and-repeat
- **Photo Mask** is **single exposure**, but less precise

Photolithography

- Lithography process refers to the formation of latent pattern on a polymer film (resist) coated on wafer surface which is then permanently transferred down to the underneath substrate through etching or implant process.

- **Patterning Layer Formation**

Done either by a growth or a deposition. This layer is to be selectively removed.

- **Photoresist Application**
- **Mask:**
- **Exposure to UV light**
- **Development and etching**

Photolithography

Photoresist

A **photoresist** is a **light-sensitive organic polymer** used to transfer a geometric pattern from a photomask onto a substrate (usually a silicon wafer) during the **photolithography** step of IC fabrication.

When exposed to **UV light**, **deep UV (DUV)**, or **EUV radiation**, the chemical structure of the photoresist changes, allowing **selective removal (development)** of either the exposed or unexposed areas.

Photoresist Composition

A typical photoresist formulation includes:

Component	Function
Polymer (Resin)	Forms the base film and provides mechanical stability
Photoactive Compound (PAC)	Sensitive to radiation; changes solubility upon exposure
Solvent	Controls viscosity; allows uniform spin coating
Additives	Improve adhesion, contrast, or thermal stability

Photolithography

- **Photoresist Application**

Feature	Positive Resist	Negative Resist
Exposure effect	Exposed region becomes soluble	Exposed region becomes insoluble
Pattern result	Mask = wafer pattern (direct)	Mask = inverse pattern (reversed)
Resolution	Higher (used for sub-100 nm features)	Lower (swelling, crosslinking blur)
Etch resistance	Lower	Higher
Applications	Mainstream IC lithography (g-line to EUV)	MEMS, packaging, thick resist, niche IC use

Key Applications

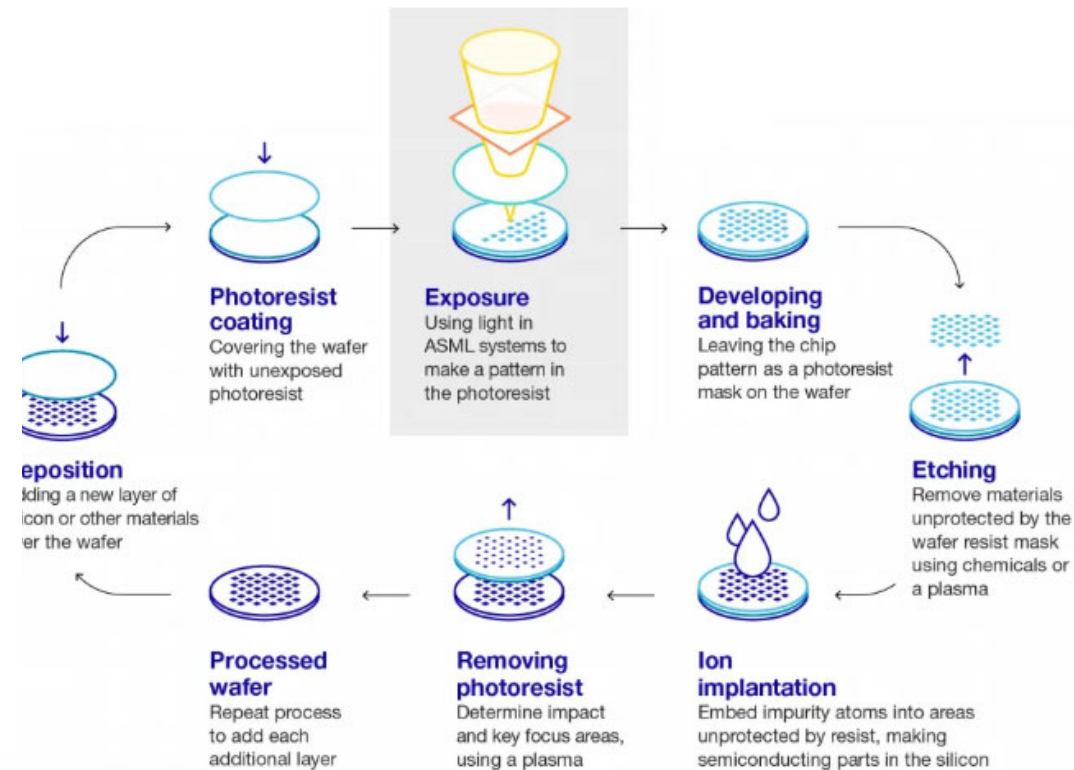
- Patterning **active regions, gates, interconnects, vias**

Photolithography

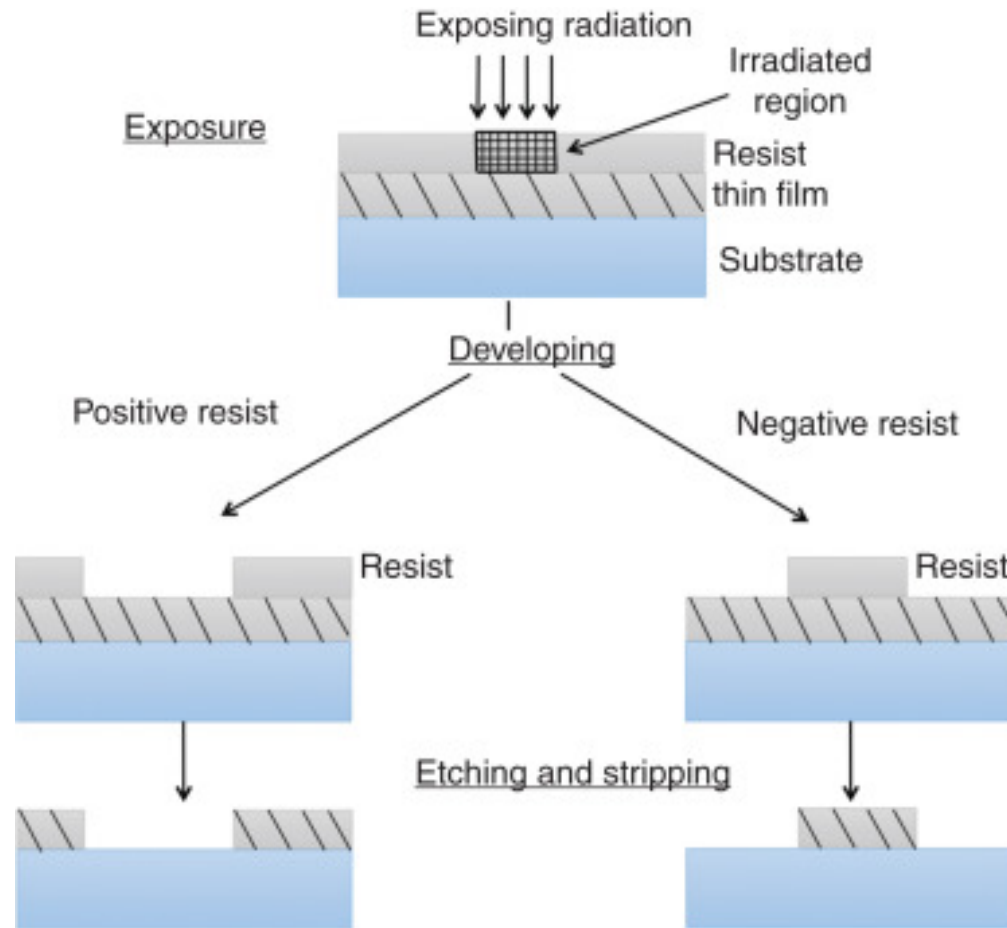
- Photoresist Application

Processing Steps Involving Photoresist

1. Surface Cleaning (remove contaminants)
2. HMDS Coating (promotes adhesion)
3. Spin Coating of Resist
4. Soft Bake (evaporate solvent)
5. Exposure (through photomask)
6. Post-Exposure Bake (in chemically amplified resists)
7. Development (using alkaline developer like TMAH)
8. Hard Bake (improve durability before etching)

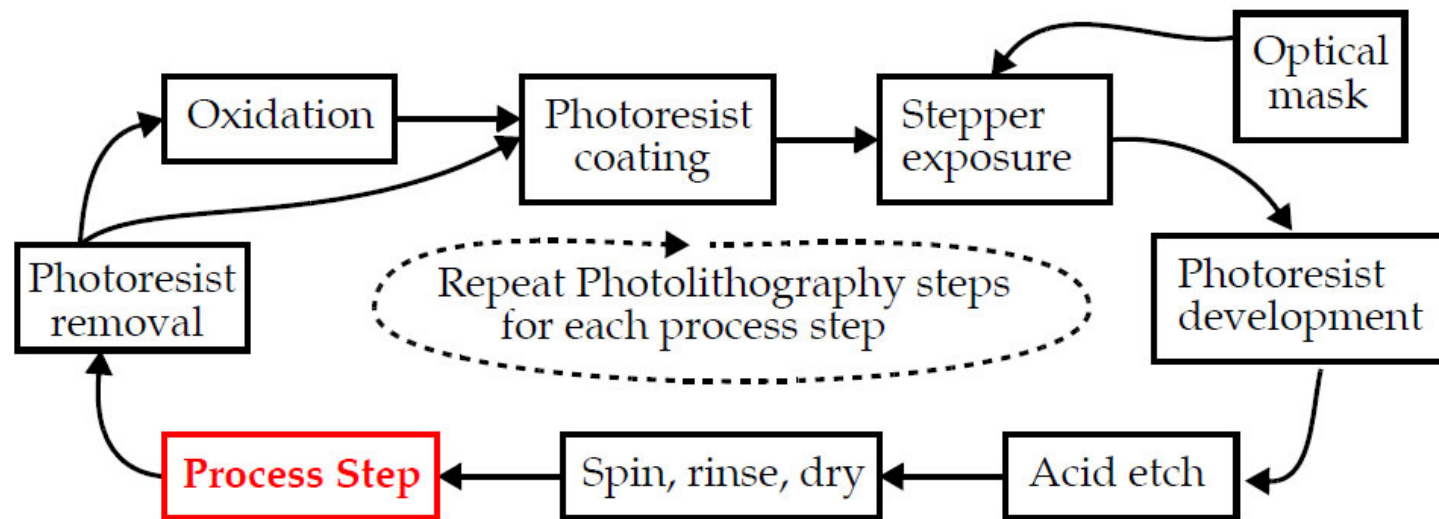


Photolithography



Ref: L.F. Thompson, An introduction to lithography, in: Introduction to Microlithography, ACS Symposium Series, Washington, DC, 1983.

Photolithography



Photolithography- Resolution

Definition of Resolution

Resolution in photolithography refers to the **smallest feature size** (minimum line width or space) that can be accurately patterned on a wafer using a given optical system and photoresist process.

It determines how **fine or dense** the circuit patterns (transistors, interconnects, etc.) can be — directly affecting the **technology node** (e.g., 180 nm, 65 nm, 5 nm, etc.).

Resolution Equation (Rayleigh Criterion)

$$R = k_1 \frac{\lambda}{NA}$$

Photolithography- Resolution

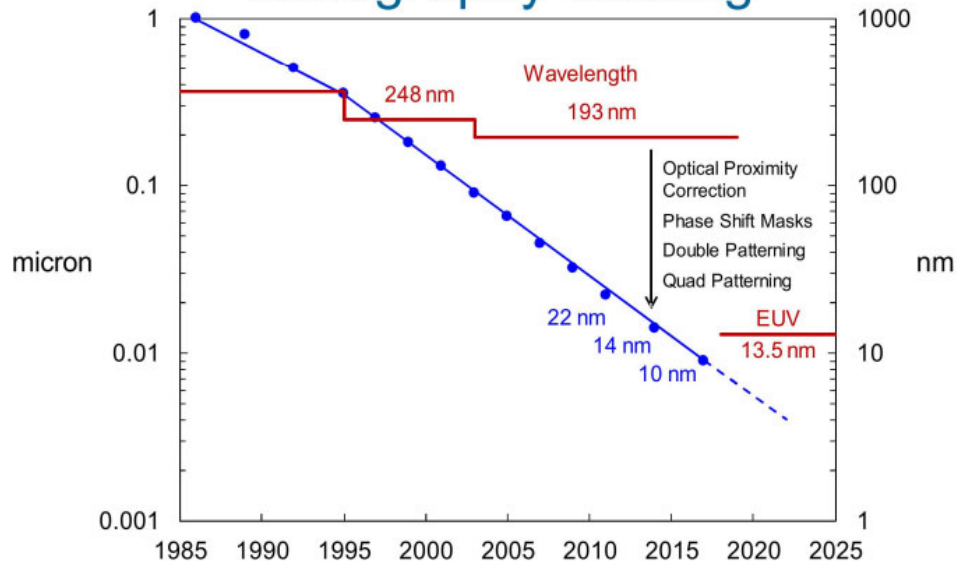
Resolution Equation (Rayleigh Criterion)

$$R = k_1 \frac{\lambda}{NA}$$

	Parameter	Description
R	Resolution	Minimum printable feature size
λ	Wavelength	Exposure light wavelength (e.g., 365 nm, 248 nm, 193 nm, 13.5 nm)
NA	Numerical Aperture	Light-gathering ability of the projection lens
k_1	Process factor	Depends on resist, illumination, and process optimization (~0.25–0.8)

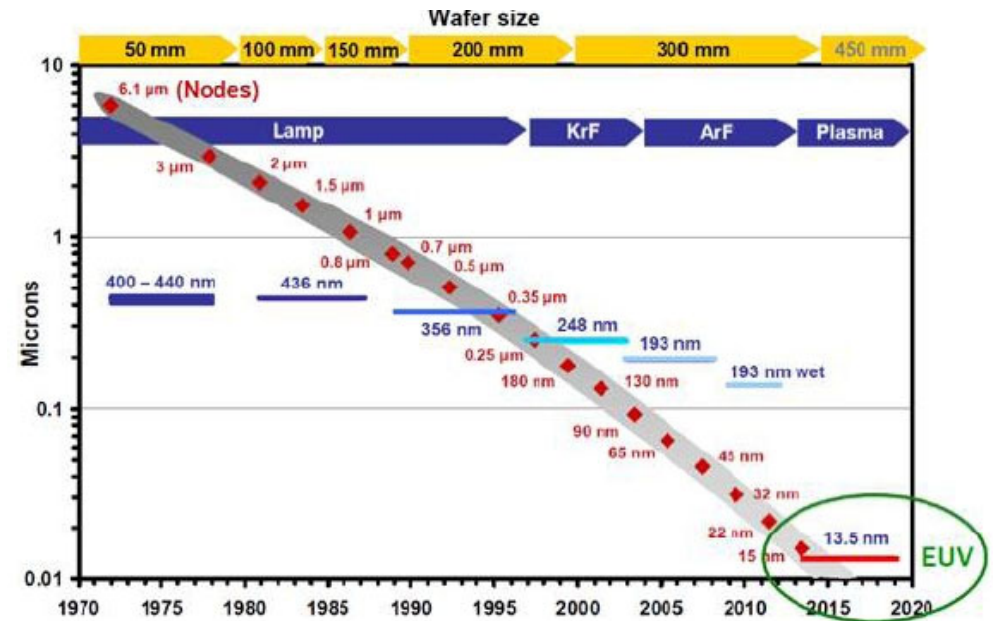
Wavelength scaling and its impacts

Lithography Scaling



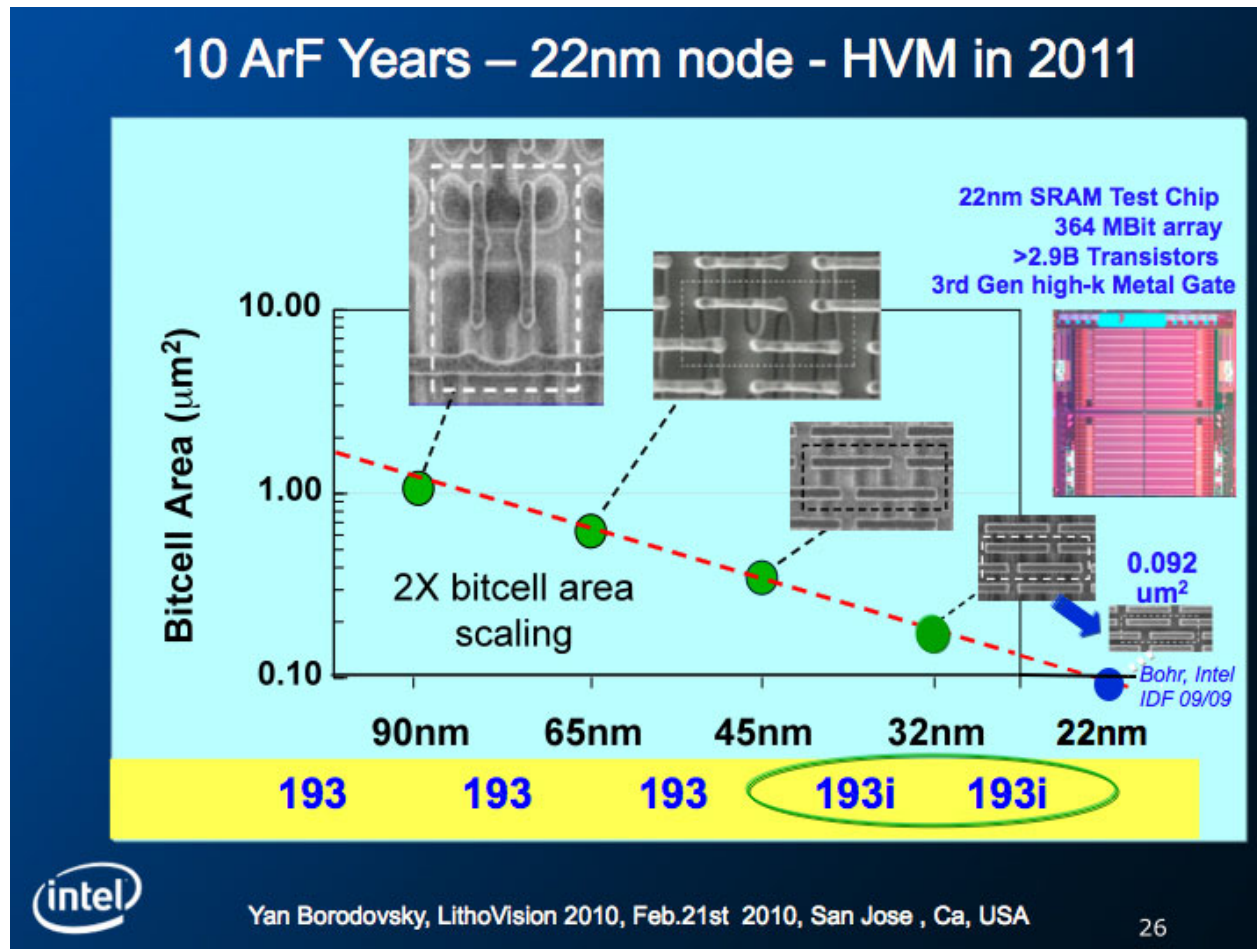
EUV will enable several more generations of scaling

$$R = k_1 \frac{\lambda}{NA}$$

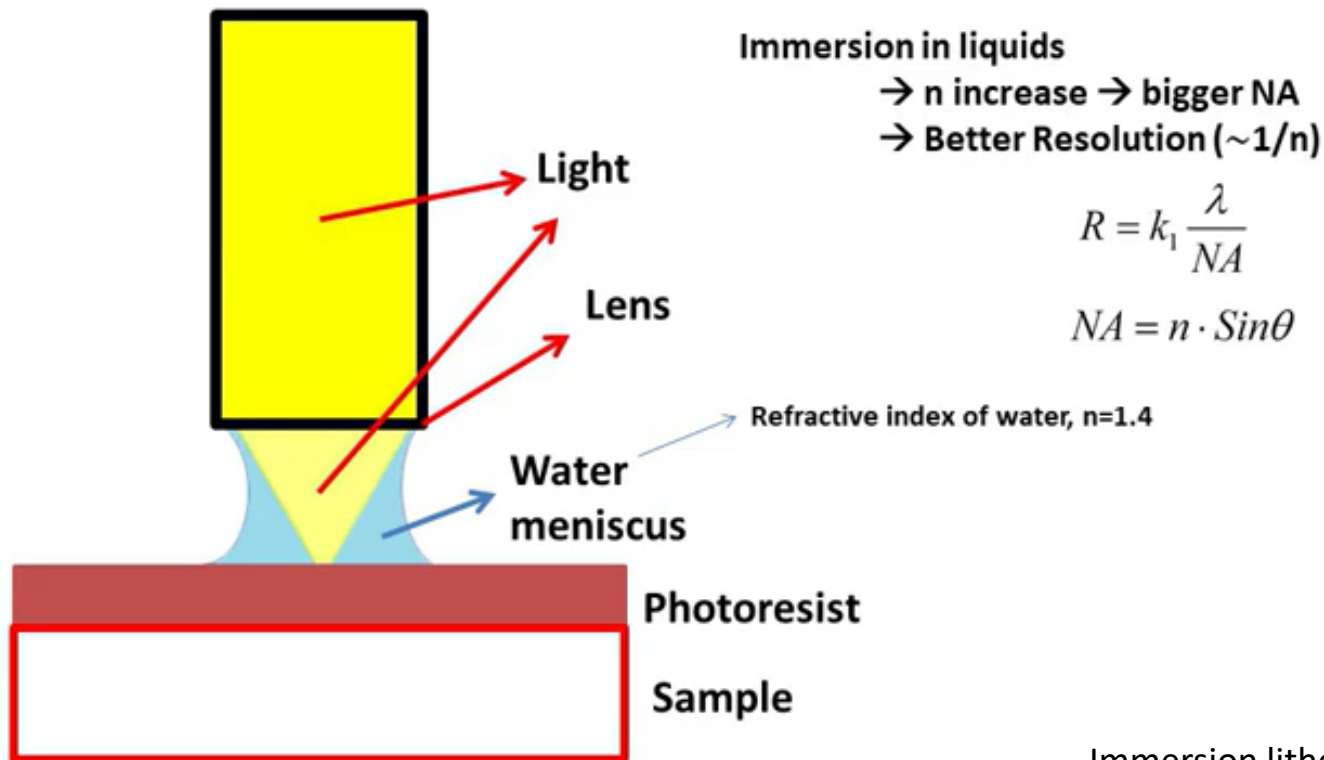


Ref: DRAM Chip market primed for long lasting resilient up cycle, 2013

193nm ArF ([Argon fluoride laser](#)) Photolithography



193nm (i) ArF Photolithography



Immersion lithography was a **key enabler** for 45 nm → 28 nm technology nodes.

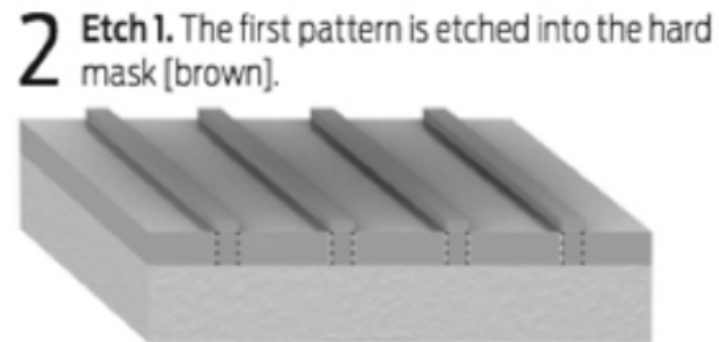
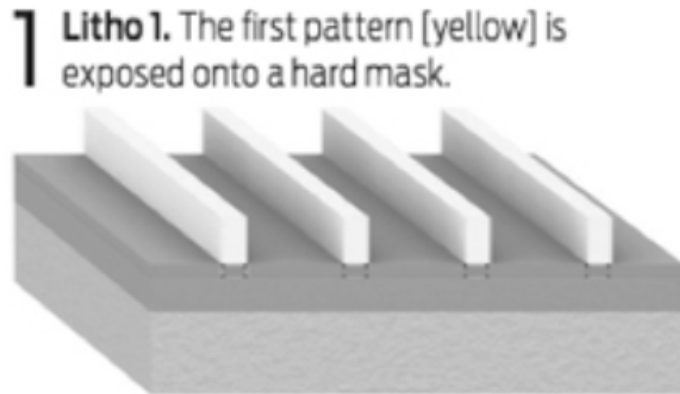
193nm ArF Photolithography-Double Patterning

- **Litho-Etch-Litho-Etch**
- **Self-Aligned Double (Multiple) Patterning**

193nm ArF Photolithography-Double Patterning

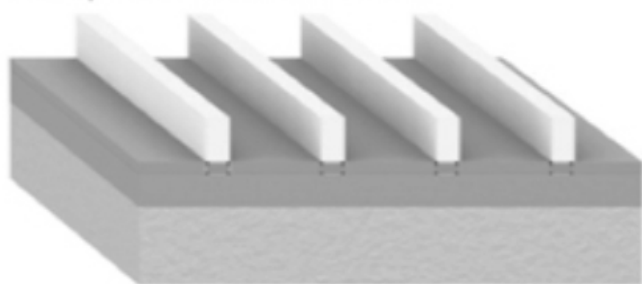
- **Litho-Etch-Litho-Etch**
 - First double patterning technique
 - Doubles the processing steps, so doubles the cost and reduces throughput
 - Requires a hard mask
 - Suffers from mask overlay issues

Litho-etch-litho-etch (LELE) is a form of double patterning. LELE is also called pitch splitting. In LELE, two separate lithography and etch steps are performed to define a single layer, thereby doubling the pattern density.

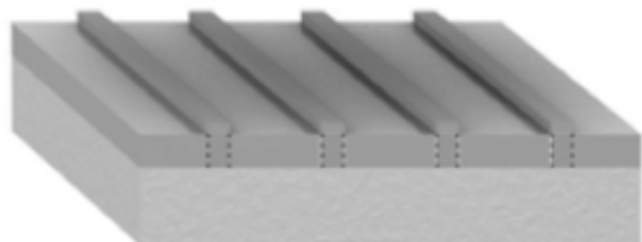


193nm ArF Photolithography-Double Patterning

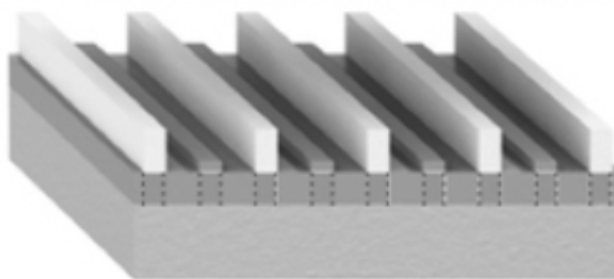
1 Litho 1. The first pattern [yellow] is exposed onto a hard mask.



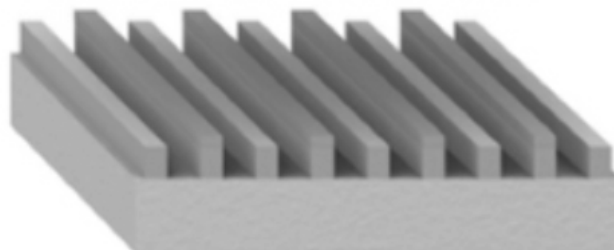
2 Etch 1. The first pattern is etched into the hard mask [brown].



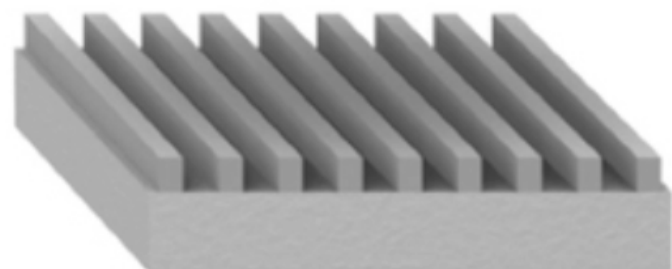
3 Litho 2. A second pattern [yellow] is exposed onto silicon [blue], doubling pattern density.



4 Etch 2. The final, double-density pattern is engraved into the silicon.

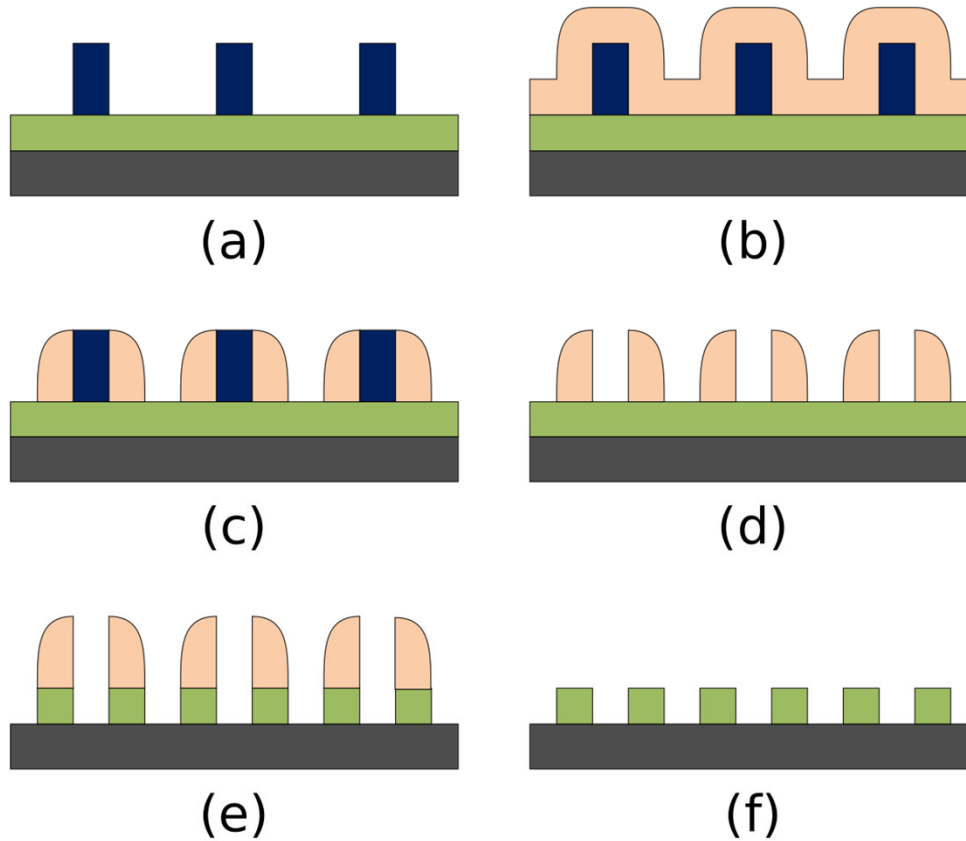


5 Wash. The remaining mask is washed away.



193nm ArF Photolithography-Double Patterning

- **Self-Aligned Double Pattering (SADP)**



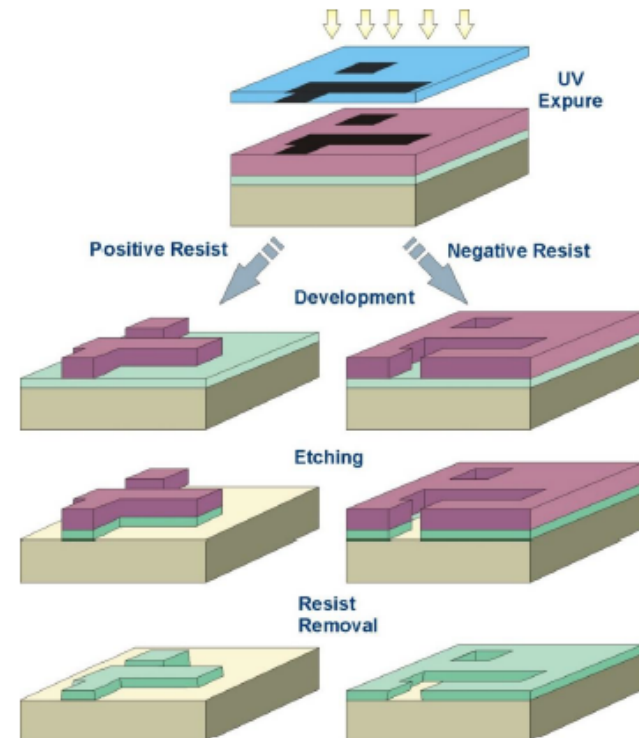
[https://commons.wikimedia.org/wiki/File:Self-aligned_\(spacer\)_double_patterning_\(SaDP\).svg](https://commons.wikimedia.org/wiki/File:Self-aligned_(spacer)_double_patterning_(SaDP).svg)

Etching (Development)

Etching is a process by which material is removed from a substrate or from thin films on the substrate surface. When a mask layer is used to protect specific regions of the wafer surface, the goal of the etching is to «precisely» remove the material which is not covered by the mask

There are two fundamental groups of etching:

- **wet etching** (liquid-based etchants)
- **dry etching** (plasma-based etchants, but not only ...)



REF:

Etching (Development)

Etching

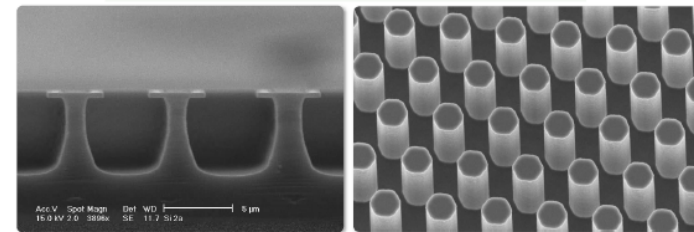
Isotropic

- etches equally in all direction
- wet etch is isotropic
- this etch leads to “undercutting”

Anisotropic

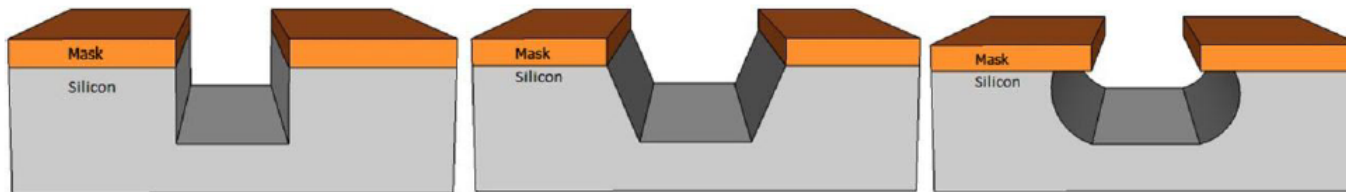
- the etch rate is dependant on the direction of
- dry etch is anisotropic

AN EXAMPLE



ISOTROPIC (LEFT) AND ANISOTROPIC (RIGHT)

Anisotropic etching has different etch rates in the lateral and vertical directions.



(a) Completely anisotropic (b) Partially anisotropic and (c) Isotropic etching of silicon

Etching (Development)

Etching

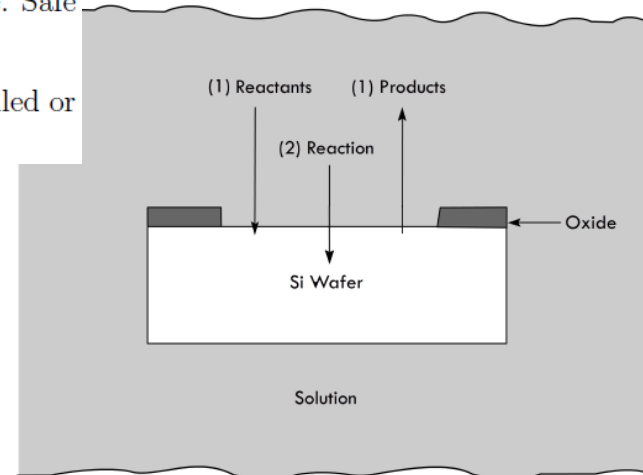
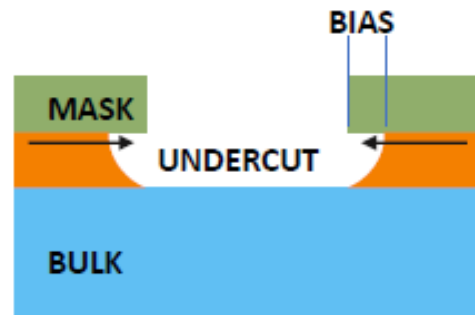
- Etching is the process of removing material from the substrate
- Etches can remove Si, SiO₂, polysilicon, and metal depending on what we want to accomplish

Etching

- there are two common types of etch processes
 - “Wet Etch” - also called Chemical Etch
 - this uses Hydrofluoric Acid (HF acid)
 - the wafer is submerged in the acid
 - simple, but produces toxic waste

Etching (Development)-Wet

1. Wet etching is used for large pattern sizes, usually larger than $2\ \mu\text{m}$.
2. It is an isotropic process - sloped sidewalls rather than straight walls.
3. Wet etch has to be combined with subsequent rinse and dry steps. This increases chances of defects or contamination.
4. Hazardous chemicals and conditions are used, so safety is an issue. Safe disposal of chemicals is essential.
5. Undercutting and resist peel off can happen if time is not controlled or etch conditions change during process.



Schematic of the wet etching process.

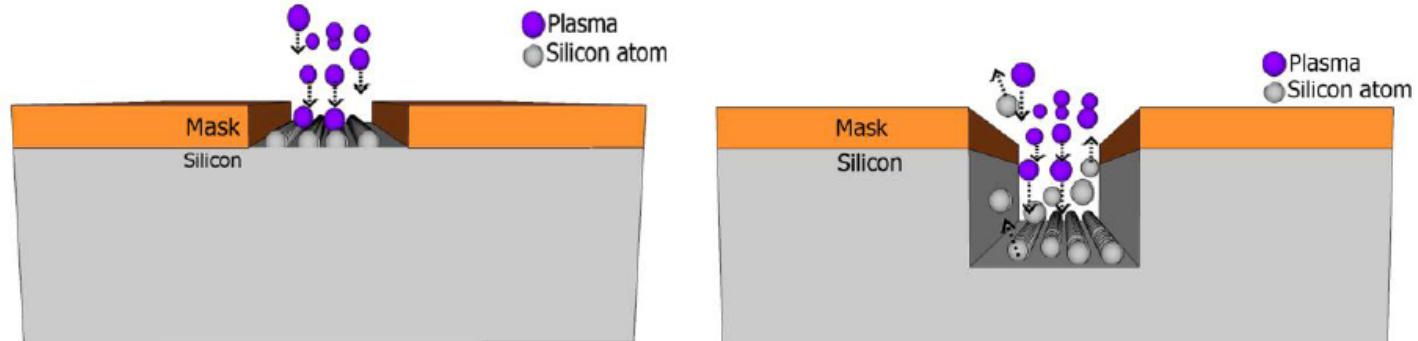
Etching (Development)

- “ Dry Etch” - also called Plasma Etch
 - also called Reactive-Ion Etch (RIE)
 - plasma is a charged gas which has excited ions (i.e., free electrons in the outer orbital)
 - plasma can be *moved* by applying an E-field
 - the chemical reaction forms a new compound that is loose and may be removed
 - since the ions move up and down, we can make a very vertical etch pattern

Etching (Development)-Dry

Physical dry etching:

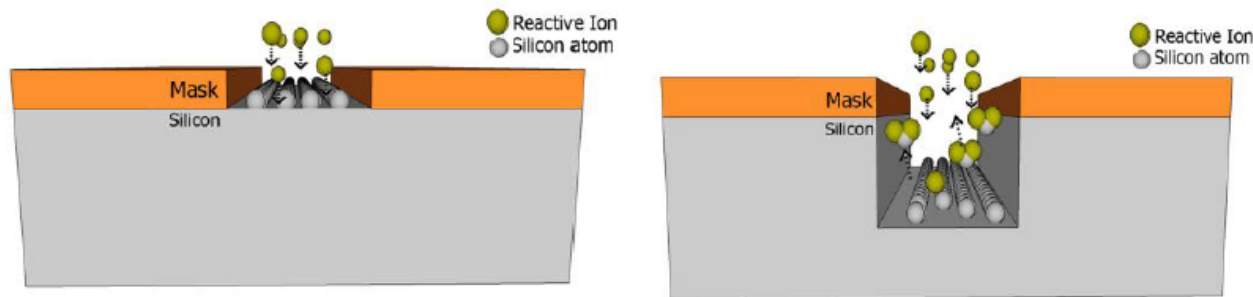
Physical dry etching requires high energy kinetic energy (ion, electron, or photon) beams to etch off the substrate atoms. When the high energy particles knock out the atoms from the substrate surface, the material evaporates after leaving the substrate. There is no chemical reaction taking place and therefore only the material that is unmasked will be removed. The physical reaction taking place is illustrated in Figure.



Etching (Development)-Dry

Chemical dry etching:

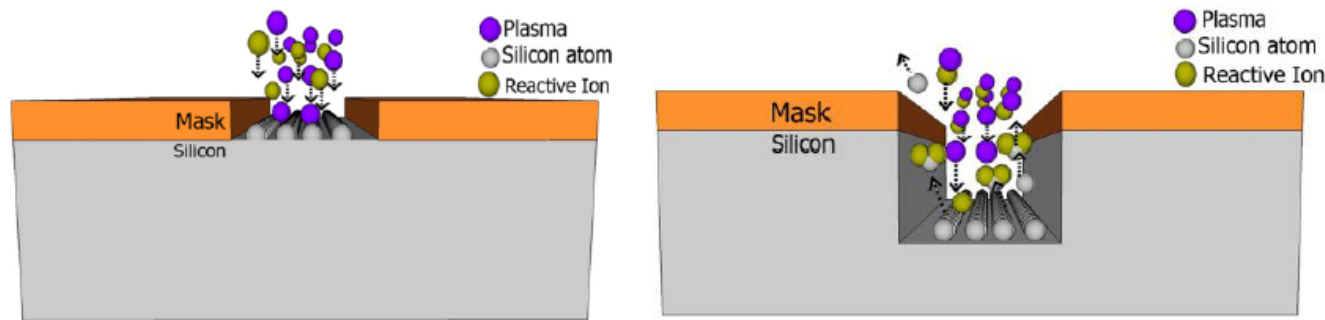
Chemical dry etching (also called vapor phase etching) does not use liquid chemicals or etchants. This process involves a chemical reaction between etchant gases to attack the silicon surface. The chemical dry etching process is usually isotropic and exhibits high selectivity. Anisotropic dry etching has the ability to etch with finer resolution and higher aspect ratio than isotropic etching. Due to the directional nature of dry etching, undercutting can be avoided.



Etching (Development)-Dry

Reactive Ion Etching:

Reactive ion etching (RIE) uses both physical and chemical mechanisms to achieve high levels of resolution. The process is one of the most diverse and most widely used processes in industry and research. Since the process combines both physical and chemical interactions, the process is much faster. The high energy collision from the ionization helps to dissociate the etchant molecules into more reactive species.



The RIE process. This process involves both physical and chemical reactions to etch off the silicon.

Wet Etch and Dry Etch

Feature	Wet Etching	Dry Etching
Medium	Liquid chemicals	Plasma (gas phase)
Etch Rate	Fast (100 – 10,000 nm/min)	Slower (10 – 1000 nm/min)
Directionality	Isotropic (poor control)	Anisotropic (good control)
Control	Moderate (affected by diffusion)	Excellent (nm-scale precision)
Damage Risk	Minimal chemical damage	Plasma damage possible
Applications	Bulk removal, MEMS, simple layers	IC nanometer-scale patterning

Etchants Applications



- Potassium hydroxide (KOH) Used to etch bulk Silicon
- Hydrofluoric Acid (HF) Used to etch Silicon Dioxide and Tungsten
- Sulfur Hexafluoride (SF₆) Used to etch Silicon Nitride Used
- Boron Trichloride (BCl₃) to etch Aluminum